

Graph Classes and Width Parameters in Lax

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This catalog follows the chapter order of the supplied survey, retaining only chapters represented by formalized definitions. Chapter numbers match the survey. Arrows link to source concepts; marked passages open their Lean encodings alongside the PDF.

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1 Basic Concepts

1.1 Basic graph notions

These references collect graph classes, their closure properties, graph parameters, and elementary graph families.

Graph classes. The sparsity lectures represent a class by its finite simple graphs on canonical vertex sets, giving a common starting point for the later sparsity notions; see [↗](#).

Isomorphism-closed graph classes. Another treatment makes closure under isomorphism explicit and provides union and intersection operations; see [↗](#).

Closure properties. This concept groups closure under induced subgraphs, minors, contractions, disjoint-union summands, and disjoint union. In particular, induced-subgraph closure is the usual hereditary property; see ↗.

Graph parameters. This concept gives a shared signature for natural-number graph parameters and records when two parameters bound one another up to functions; see ↗.

Paths. These are the finite path graphs, the simplest connected examples in the list; every path is a tree. See ↗.

Stars. These trees have one centre joined to all other vertices; their leaves have no edges between them. See ↗.

Trees. A tree is connected and acyclic, so both paths and stars form subclasses of the tree family. See ↗.

Ladders. A ladder is a two-row grid, with two paths joined by corresponding rungs. See ↗.

1.2 Some special graphs

The remaining examples follow the survey's special-graph discussion. Where the classes are nested, triangles lie among maximal outerplanar graphs, and outerplanar graphs lie among planar graphs. Grids, walls, series-parallel graphs, and triangulations provide further examples in this chapter.

Triangles. This is the three-vertex complete graph, which is also a maximal outerplanar graph. See ↗.

Maximal outerplanar graphs. These outerplanar graphs are maximal under adding edges while preserving outerplanarity; every triangle is an example. See ↗.

Outerplanar graphs. They admit a crossing-free drawing with every vertex on the outer face, so this family is contained in the planar graphs. See ↗.

Series-parallel graphs. This class is generated from a single edge by two-terminal series and parallel constructions. See ↗.

Triangulations. These are maximal planar graphs, obtained by triangulating the faces of a planar embedding. See ↗.

Grids and walls. Rectangular grids and their brick-wall variants are standard planar examples and recurring structural patterns. See ↗.

Planar graphs. These graphs admit a drawing in the plane with no crossing edges; the formalized entry records a straight-line drawing as a certificate. See ↗.

2 Perfection, Generalized Perfection, and Related Concepts

2.1 Related concepts

χ -boundedness. This class property bounds chromatic number as a function of clique number, a control closely related to graph perfection. See ↗.

3 Cycles, Chords, and Bridges

3.1 Cycle deletion parameters

These parameters count the deletions needed to remove all cycles, by deleting vertices or edges.

Feedback vertex number. This measures the minimum number of vertices to remove so the remaining graph is a forest. See ↗.

Feedback edge number. This is the edge-deletion counterpart: remove edges until the remaining graph is acyclic. See ↗.

5 Vertex and Edge Orderings

5.1 Orderings and neighborhood measures

The survey treats elimination and neighborhood-based orderings; these formalized concepts likewise use linear orders and reachability.

Admissibility. Relative to an ordering, this parameter counts short paths from a vertex to earlier vertices that meet only at their start. See ↗.

Generalized coloring numbers. Weak and strong reachability count vertices accessible by short paths under a linear order; the parameters minimize the largest count over all orders. See ↗.

Almost-linear neighborhood complexity. This bounds the number of distinct neighborhood traces on a vertex set by a near-linear function of its size. See ↗.

Linear neighborhood complexity. This companion formulation asks for a linear bound on distinct neighborhood traces. See ↗.

7 Forbidden Subgraphs

7.1 Forbidden induced subgraphs

Cographs. Their forbidden-pattern characterization uses the four-vertex path; this is the same class referenced in the perfection chapter. See ↗.

Induced minors. This containment variant preserves adjacency and non-adjacency between retained branch sets. See ↗.

Excluded induced grid minors. This class excludes an induced minor isomorphic to a square grid of the specified size. See ↗.

Weakly sparse classes. The formalized property excludes a fixed complete bipartite graph as a subgraph. See ↗.

7.2 Forbidden minor characterizations

Graph minors. Branch sets package vertex deletion, edge deletion, and contraction into one containment relation. See ↗.

Topological minors. Edges of the smaller graph are represented by internally disjoint paths. See ↗.

Grids and walls. These planar structures also serve as canonical patterns in minor containment. See ↗.

Outerplanar graphs. The drawing-based class also appears in the survey's discussion of planar structure. See ↗.

Planar graphs. Their forbidden-minor characterization is a central example in this chapter. See ↗.

Triangulations. These provide maximal planar examples alongside the planarity and minor references above. See ↗.

9 Matrices and Polyhedra

9.1 Matrix-based graph parameters

Mixed minor number. This parameter detects fully mixed blocks in ordered adjacency matrices and is comparable to twin-width. See [↗](#).

10 Distance Properties

10.1 Distances, shallow minors, and sparse classes

These concepts use bounded-radius reachability or shallow containment to describe structural sparsity.

Uniform quasi-wideness. Large vertex sets contain large distance-separated subsets after removing a bounded-size separator, uniformly over the class. See [↗](#).

Shallow-minor density. This records edge density after contractions of bounded depth, for each fixed radius. See [↗](#).

Shallow topological minors. This bounds the subdivision depth of paths representing edges in a shallow topological minor. See [↗](#).

Nowhere-dense graph classes. At each fixed depth, some clique is excluded as a shallow minor of every graph in the class. See [↗](#).

Monadically dependent classes. These classes cannot produce every graph through the specified first-order transductions with vertex colorings. See [↗](#).

11 Algebraic Compositions and Recursive Definitions

11.1 Trees, k -trees, and partial k -trees

Trees. The recursively describable tree family appears here again alongside the basic graph reference. See ↗.

Treewidth. This parameter is defined by tree decompositions and their largest bag size. See ↗.

11.2 Series-parallel graphs

Series-parallel graphs. This class is generated by two-terminal series and parallel composition. See ↗.

11.3 Cographs and width parameters

Cographs. The recursive and forbidden-path views of this class are linked by the survey; its formalized entry also occurs in the earlier chapters. See ↗.

Cliquewidth. A k -expression constructs a graph using k labels; bounded cliquewidth organizes classes by the number of labels needed. See ↗.

Merge-width. This width parameter measures merge sequences using radius-dependent neighborhoods in partially resolved graphs. See ↗.

Twin-width. This partition-sequence parameter counts red neighbors created as vertex parts merge, and applies to both sparse and dense classes. See ↗.

12 Decompositions and Cutsets

12.1 Graph decompositions and width

Treewidth. Tree decompositions cover vertices and edges while keeping each vertex's bags connected; their width is measured by the largest bag. See ↗.

Twin-width. The sequence of merged vertex parts gives a second decomposition-based width measure. See ↗.